

Team name:	Lightning
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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Forint Ft	248 794,01 Ft	\$707,20

#	Component	Part name	Autor	Source	Quantity	Unit cost (Local Currency)	Unit cost (US Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)
1	Robot shell	Custom made	Cs. Kiss Zsófia	3D Printer	1	8 000,00 Ft	\$22,74	8 000,00 Ft	\$22,74
2	Main Controller	Arduino Mega 2560	ARDUINO	https://malnapc.hu/arduino-mega2560-rev3-a000067?keyword=arduino%20mega	1	19 990,00 Ft	\$56,82	19 990,00 Ft	\$56,82
3	Microcontroller	Arduino Micro	ARDUINO	https://malnapc.hu/arduino-micro-a000053?keyword=arduino%20micro	1	9 990,00 Ft	\$28,40	9 990,00 Ft	\$28,40
4	Dual Motor Driver	Dual VNH5019 motor shield	Pololu	https://www.pololu.com/product/2507	1	21 090,41 Ft	\$59,95	21 090,41 Ft	\$59,95
5	Drive Motors	DC motor 12V	Pololu	https://www.tme.eu/hu/details/pololu-3481/dc-motorok/pololu/195-1-metal-gearmotor-20dx44l-mm-12v-cb/	4	12 410,00 Ft	\$35,28	49 640,00 Ft	\$141,10
6	Wheels	Custom made	Szél Nikolett, Szilágyi Jázmin	Self Made	4	1 000,00 Ft	\$2,84	4 000,00 Ft	\$11,37
7	Voltage Regulator	5V 2.5A STEP-DOWN D24V22F5 POLOLU	Pololu	https://www.tme.eu/hu/details/pololu-2858/konverter-modulok/pololu/5v-2-5a-step-down-d24v22f5/	2	6 297,00 Ft	\$17,90	12 594,00 Ft	\$35,80

8	TOF Sensor	VL53L0X	STMicroelectronics	https://modulshop.hu/vl53l0x-tof-lezeres-tavolsagmero-szenzor-3-100-cm-771?gad_source=1&gad_campaignid=21423119443&gbraid=0AAAAAoJLZNGVge82qVGBntJjZfwxjYH5&gclid=Cj0KCQjwjdTCBhCLARIsAEu8bpLR8UHQsggVxbyDnNivsa4DX34K_nEDNU8hAmQkWPv4C5vMo6g29x0aAgu4EALw_wcB	11	1 350,00 Ft	\$3,84	14 850,00 Ft	\$42,21
9	Color Sensor	TCS34725	Adafruit	https://malnapc.hu/a/1334-rgb-color-sensor-with-ir-filter-and-white-led	1	3 490,00 Ft	\$9,92	3 490,00 Ft	\$9,92
10	Camera	OpenMV h7	OpenMV	https://openmv.io/collections/cameras/products/openmv-camera-h7-r2?variant=39399410106462	2	28 144,00 Ft	\$80,00	56 288,00 Ft	\$160,00

11	OLED Display	SSD1306	Solomon Systech	https://modulshop.hu/096-oled-kijelzokek-12864-i2c-733?gad_source=1&gad_campaignid=21423119443&gbraid=0AAAAAoJLZNGVge82qVGBntJjZfwfxjYH5&gclid=Cj0KCQjwjdTCBhCLARIsAEu8bpJrSz6AGHqoCmJ4NyY1XRamDJSTOzT6EMuhn6tgnOceBWaHkJERbZcaAI42EALw_wcB	1	1 400,00 Ft	\$3,98	1 400,00 Ft	\$3,98
12	Gyro Sensor	MPU6050	InvenSense	https://www.hestore.hu/prod_10035607.html?gad_source=1&gad_campaignid=21163162680&gbraid=0AAAAADz0y9lULjZHQt1B0Ko7pgoeVCTr3&gclid=Cj0KCQjwjdTCBhCLARIsAEu8bpLFTs4v1XzVJc3mc0YjmkqQM6jAlqwW0kDdMRYaZ8zwvpKXzgVUIUaAqfiEALw_wcB	1	1 246,00 Ft	\$3,54	1 246,00 Ft	\$3,54
13	Infra Sensor	QTRXL	Pololu	https://www.tme.eu/hu/details/pololu-4601/tavolsag-erzekelok/pololu/qtrxl-hd-01a-reflectance-sensor/	1	1 256,00 Ft	\$3,57	1 256,00 Ft	\$3,57
14	Lipo Battery	1350 mAh LiPo 7.4V-20C	Sunpadow	https://www.aliexpress.com/	1	10 554,00 Ft	\$30,00	10 554,00 Ft	\$30,00

15	Lipo Battery	1100 mAh LiPo 11.1V-35C	Sunpadow	https://www.aliexpress.com/	1	11 257,60 Ft	\$32,00	11 257,60 Ft	\$32,00
16	Servo for dispenser system	Metal Gear Servo	DFROBOT	https://www.tme.eu/hu/details/df-ser0046/szervomotorok/dfrobot/ser0046/	1	3 148,00 Ft	\$8,95	3 148,00 Ft	\$8,95
17	Connecting parts	Screws, washers, nuts, spacers, Shrink tube		https://www.tme.eu/hu/	1	10 000,00 Ft	\$28,43	10 000,00 Ft	\$28,43
18	Circuit components	LED, switch, wires...		https://www.tme.eu/hu/	1	10 000,00 Ft	\$28,43	10 000,00 Ft	\$28,43

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Forint Ft	0,00 Ft	\$0,00

#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local Currency)	Cost (USA Dollars)
1	VL53L0X-Arduino	Library	For controlling the VL53L0X distance sensors on Arduino	https://github.com/triantara/ARDUINO-TRIANTARA/blob/main/VL530X%20multi.zip	triantara	FREE	FREE
2	MPU6050-Arduino	Library	For controlling MPU6050 on Arduino	https://arduinoprojekt.wordpress.com/accelerometer-mpu-6050/	Krodal	FREE	FREE
3	DualVHN5019 motor shield	Library	For controlling the motor shield on Arduino	https://github.com/pololu/dual-vnh5019-motor-shield	RyanTM (POLOLU)	FREE	FREE
4	OLED display	Library	For controlling the OLED display on Arduino	https://github.com/adafruit/Adafruit_SSD1306	makermelissa	FREE	FREE
5	Servo	Library	For controlling the servo motor on Arduino	https://docs.arduino.cc/libraries/servo/	ARDUINO	FREE	FREE
6	TCS34725	Library	For controlling the color sensor on Arduino	https://github.com/adafruit/Adafruit_TCS34725	Tyeth	FREE	FREE
7	IDE Arduino	Software's Tool	Software	https://www.arduino.cc/en/software/	Ardiono.cc	FREE	FREE

8	OpenMV IDE	Software's Tool	Software	https://openmv.io/pages/download?srsltid=AfmBOorCBEDHRSNbN5vIHPB5Pp8ZWOH_pafGGo0tPwz1zhkNicLEXu4o	OpenMV	FREE	FREE
9	FreeCAD	Software's Tool	Software	https://www.freecad.org/downloads.php	The FreeCAD Team	FREE	FREE